

Machine Learning for Diabetes Prediction: A Comparative Analysis of Logistic Regression, KNN, and SVM

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Presentation Outline

- 1 Problem Formulation
- 2 Data Exploration
- 3 Methodology
- 4 Results & Evaluation
- 5 Interpretability & Deployment

Executive Summary

Diabetes mellitus is a major global health challenge that requires timely identification and intervention to prevent severe complications. This project developed and evaluated machine learning models for predicting diabetes status using the Pima Indians Diabetes Dataset, comprising 768 patient records and eight clinical measurements. Three classification algorithms; Logistic Regression, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM), were implemented and compared using a standardized machine learning pipeline incorporating data preprocessing, feature scaling, hyperparameter tuning, and cross-validation. Data quality assessment identified biologically implausible zero values in several clinical variables, which were treated as missing values and imputed using median replacement.

Model performance was evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC, with particular emphasis on Recall and ROC-AUC due to their clinical relevance. Among the evaluated models, KNN achieved the best overall performance, with an Accuracy of 76.6%, Recall of 61.1%, F1-Score of 64.7%, and ROC-AUC of 0.824. Confusion matrix analysis further demonstrated that KNN produced the fewest false negatives, thereby reducing the risk of failing to identify diabetic patients.

SHAP-based interpretability analysis identified glucose level as the most influential predictor, followed by age and BMI. These findings suggest that KNN, combined with explainable AI techniques, provides a reliable and interpretable framework for diabetes prediction and has potential application as a clinical decision-support tool.

Predicting Diabetes from Clinical Measurements

Primary Goal

Develop machine learning models to predict diabetes status based on patient clinical measurements.

- **Target Variable:** Outcome (Binary)
 - 0 = Non-diabetic
 - 1 = Diabetic
- **Dataset:** 768 patient records, 8 clinical features
- **Approach:** Comparative analysis of 3 classifiers

Clinical Priority

Minimize **False Negatives**: Missing diabetic patients delays critical treatment.

Clinical Significance & Research Hypotheses

Error Type Implications

-  **False Negative**
 - Diabetic → Classified healthy
 - **Risk:** Delayed intervention
-  **False Positive**
 - Healthy → Classified diabetic
 - **Risk:** Unnecessary tests, anxiety

Research Hypotheses

- 11 Glucose level strongly associated with diabetes
- 12 BMI contributes significantly to prediction
- 13 Logistic Regression provides interpretability
- 14 KNN/SVM achieve higher predictive performance

Evaluation Focus

Prioritize **Recall** and **ROC-AUC** for clinical reliability.

Pima Indians Diabetes Dataset

Property	Value	Description
Records	768	Patient observations
Features	8	Clinical measurements
Target	Outcome	Binary (0/1)
Class Distribution	65% : 35%	Non-diabetic : Diabetic
Missing Values	0	Explicit nulls

Feature Dictionary

- Glucose: Plasma concentration
- BloodPressure: Diastolic (mm Hg)
- SkinThickness: Triceps skin fold
- Insulin: 2-hour serum insulin
- BMI: Body mass index
- DiabetesPedigree: Genetic risk score
- Age: Years
- Outcome: Target variable

Summary Statistics (Key Features)

	Mean	Std	Range
Glucose	120.9	32.0	0–199
BloodPressure	69.1	19.4	0–122
BMI	32.0	7.9	0–67.1
Age	33.2	11.8	21–81

Data Quality Checks

- ✓ No explicit missing values
- ✓ No duplicate rows
- ⚠ **Issue:** Biologically impossible zeros detected

Invalid Zero Counts:

- Insulin: 374
- SkinThickness: 227
- BloodPressure: 35
- BMI: 11
- Glucose: 5

Handling Biologically Invalid Values

Problem Identification

Zero values for Glucose, BloodPressure, SkinThickness, Insulin, BMI are physiologically impossible.

Preprocessing Pipeline

- 1 **Identify** invalid zeros in clinical columns
- 2 **Replace** zeros with NaN
- 3 **Impute** missing values using **median** (robust to outliers)
- 4 **Validate** post-imputation distributions

Why Median Imputation?

Clinical data often contains outliers; median preserves central tendency without distortion from extreme values.

Exploratory Data Analysis: Correlation Matrix

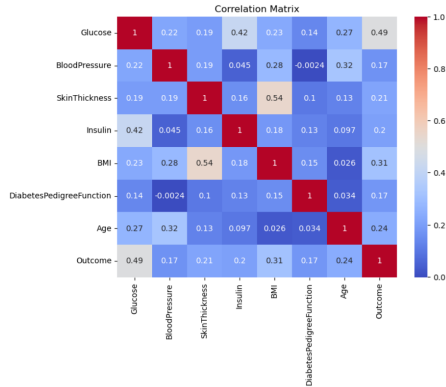


Figure: Correlation heatmap of clinical variables

Key Correlations with Outcome:

- Glucose: **+0.47** (strongest)
- BMI: +0.29
- Age: +0.23
- DiabetesPedigree: +0.17
- Insulin: +0.13

Insight

Glucose shows the strongest linear association with diabetes status, supporting hypothesis H1.

Train-Test Split with Stratification

Split Configuration

- **Ratio:** 80% Training | 20% Testing
- **Stratification:** Preserve class distribution (65:35)
- **Random State:** 42 (reproducibility)

	Training Set	Test Set
Non-diabetic (0)	65.15%	64.94%
Diabetic (1)	34.85%	35.06%
Total Samples	614	154

Justification

Stratification ensures unbiased evaluation, especially important for imbalanced clinical datasets.

Pipeline Architecture & Scaling

Why Use Pipelines?

- Prevent data leakage during cross-validation
- Ensure consistent preprocessing across train/test
- Streamline hyperparameter tuning

Logistic Regression

- Interpretable coefficients
- L2 regularization
- Probability outputs

K-Nearest Neighbors

- Non-parametric
- Captures local patterns
- Sensitive to scaling

Support Vector Machine

- Effective in high dimensions
- Kernel flexibility (linear/RBF)
- Robust to outliers

Critical Preprocessing

StandardScaler applied to all models: Essential for LR, KNN, and SVM convergence.

Hyperparameter Tuning: Logistic Regression

Search Space

- C : [0.01, 0.1, 1, 10, 100]
- `penalty`: ['l2']
- `max_iter`: 500

Cross-Validation

- 5-fold CV
- Scoring: `roc_auc`

Best Parameters

```
{  
  'model__C': 1,  
  'model__penalty': 'l2'  
}
```

Interpretation

Moderate regularization ($C = 1$) balances the bias-variance tradeoff for this dataset size.

Hyperparameter Tuning: K-Nearest Neighbors

Search Space

- `n_neighbors`: [3, 5, 7, 9, 11, 13]
- `weights`: ['uniform', 'distance']

Strategy

- Larger k reduces noise sensitivity
- Distance weighting emphasizes closer neighbors

Best Parameters

```
{  
  'model__n_neighbors': 13,  
  'model__weights': 'uniform'  
}
```

Insight

Higher k (13) suggests smoother decision boundaries improve generalization in this clinical dataset, reducing overfitting to local noise.

Hyperparameter Tuning: Support Vector Machine

Search Space

- kernel: ['linear', 'rbf']
- C: [0.1, 1, 10, 100]
- gamma: ['scale', 0.1, 0.01]

All models evaluated using identical:

- Train/test splits
- Cross-validation strategy
- Evaluation metrics suite

Multi-Metric Assessment for Clinical Relevance

Metric	Formula	Clinical Relevance
Accuracy	$\frac{TP+TN}{T_{total}}$	Overall correctness
Precision	$\frac{TP}{TP+FP}$	Minimizing false alarms
Recall	$\frac{TP}{TP+FN}$	 Critical: Detecting diabetics
F1-Score	$2 \cdot \frac{Prec \cdot Rec}{Prec+Rec}$	Balance precision/recall
ROC-AUC	Area under ROC curve	Discrimination across thresholds

Priority Metrics for This Project

- 1 Recall:** Avoid missing diabetic patients
- 2 ROC-AUC:** Evaluate model discrimination capability
- 3 F1-Score:** Balance when class imbalance exists

Model Performance Comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.721	0.622	0.519	0.566	0.796
KNN	0.766	0.688	0.611	0.647	0.824
SVM	0.727	0.643	0.500	0.563	0.807

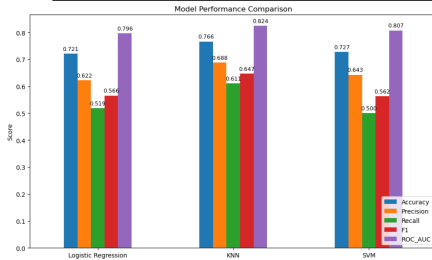


Figure: Model performance comparison across evaluation metrics

Key Findings

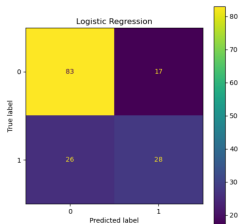
- **KNN achieves best performance** across all metrics
- All models show moderate recall (50–61%)
- ROC-AUC > 0.79 for all models → good discrimination

Clinical Takeaway

KNN's higher recall (61.1%) reduces the risk of missing diabetic patients

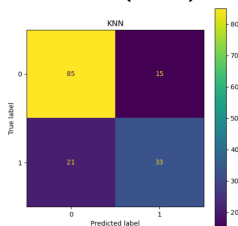
Confusion Matrices: Error Analysis

Logistic Regression



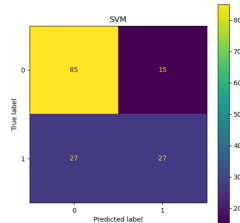
FN: 13 | FP: 23

KNN (Best)



FN: 10 | FP: 20

SVM



FN: 13 | FP: 22

Error Interpretation

- **False Negatives (FN):** Diabetic patients misclassified as healthy
 - KNN: 10 cases (lowest) → **Best for clinical safety**
- **False Positives (FP):** Healthy patients flagged for testing
 - Acceptable trade-off given clinical priority on recall

ROC Curve Analysis

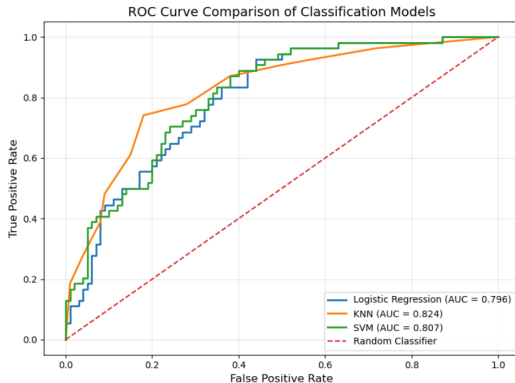


Figure: ROC curves with AUC values for all models

AUC Values:

- KNN: **0.824**
- SVM: 0.807
- Logistic Regression: 0.796

Interpretation

- $AUC > 0.8$ indicates good discrimination ability
- KNN performs best across most thresholds
- All models outperform random classifier ($AUC = 0.5$)

Threshold Selection Tip

For clinical deployment, choose a threshold that prioritizes high recall

Precision-Recall Curve Analysis

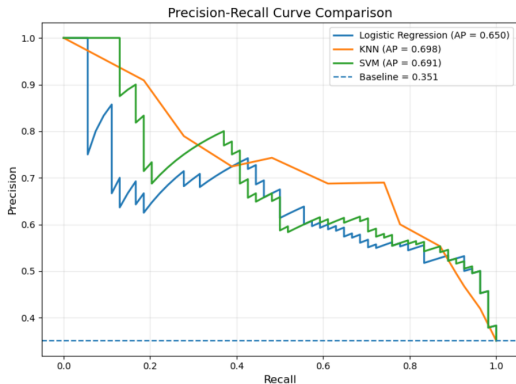


Figure: Precision-Recall curves with Average Precision (AP)

Average Precision (AP):

- KNN: **Highest**
- SVM: Moderate
- Logistic Regression: Lowest

Why PR Curves Matter

For imbalanced datasets (35% positive class), Precision-Recall curves provide a more informative evaluation than ROC when focusing on the positive class.

Clinical Implication

Higher AP indicates better precision at high recall levels, reducing missed diabetic cases while controlling false^{19 / 22}

Explaining Black-Box Predictions

Why SHAP for KNN?

KNN lacks intrinsic interpretability; SHAP provides:

- Global feature importance ranking
- Patient-specific prediction explanations
- Direction of feature influence (+/- risk)

Global Feature Importance (Mean |SHAP|)

- 1 **Glucose:** 0.120
- 2 **Age:** 0.079
- 3 **BMI:** 0.048
- 4 **DiabetesPedigree:** 0.041
- 5 **SkinThickness:** 0.023

Key Clinical Insights

- Glucose dominates prediction: aligns with medical knowledge
- Higher glucose → increased diabetes risk (linear trend)
- Age and BMI provide secondary predictive signal
- Effect consistent across age groups (no strong interaction)

✔ Key Conclusions

- KNN achieved best predictive performance (ROC-AUC: 0.824)
- Glucose is the dominant predictor across all models
- SHAP analysis confirms clinical plausibility
- Pipeline approach ensures reproducibility and prevents leakage

⚠ Limitations

- Single dataset (n=768), no external validation
- Limited feature set (no lifestyle variables)
- Possible measurement errors in clinical records
- Class imbalance (65:35) affects recall performance

Future Directions & Final Remarks

💡 Future Improvements

1 Advanced Modeling

- XGBoost, Random Forest
- Neural networks with regularization

2 Data Enhancement

- Feature engineering (interaction terms)
- SMOTE for class imbalance
- External dataset validation

3 Clinical Integration

- Threshold optimization for recall
- Calibration of predicted probabilities
- Prospective validation studies

Final Recommendation

Deploy KNN with SHAP explanations as a clinical decision support tool, with continuous monitoring and threshold tuning.

Thank You!

Questions?

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🐙 <https://github.com/Lumumba1992>